Software Development Models ? Waterfall, Agile, Spiral & V-Model

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Comparative Paradigm Analysis

Software engineering methodologies span a continuum from predictive (plan-driven) to adaptive (change-driven) frameworks. Selection depends on requirements stability, safety criticality, project duration, and stakeholder availability.

2. Predictive Models: Waterfall & V-Model

? Waterfall: Linear-sequential progression where each phase must complete and receive formal sign-off before subsequent phases begin. Best suited for high-assurance firmware, medical devices, and aerospace systems with immutable specifications.

? V-Model (Verification & Validation): Maps each development phase directly to a corresponding testing tier. Requirements correspond to User Acceptance Testing (UAT); System Design corresponds to System Testing; Detailed Design corresponds to Integration Testing; Coding corresponds to Unit Testing.

3. Iterative & Risk-Driven: Spiral Model

Proposed by Barry Boehm, the Spiral Model organizes development into repeating cycles across four quadrants: Objective Setting, Risk Assessment & Resolution, Development & Verification, and Review/Planning. Each cycle produces a refined prototype, making it optimal for complex, high-uncertainty R&D projects.

4. Adaptive Frameworks: Agile & Scrum

Agile prioritizes individuals and interactions, working software, customer collaboration, and responsiveness to change. Scrum operationalizes Agile through time-boxed Sprints (1 to 4 weeks), continuous backlog grooming, daily stand-up ceremonies, and sprint retrospectives.

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